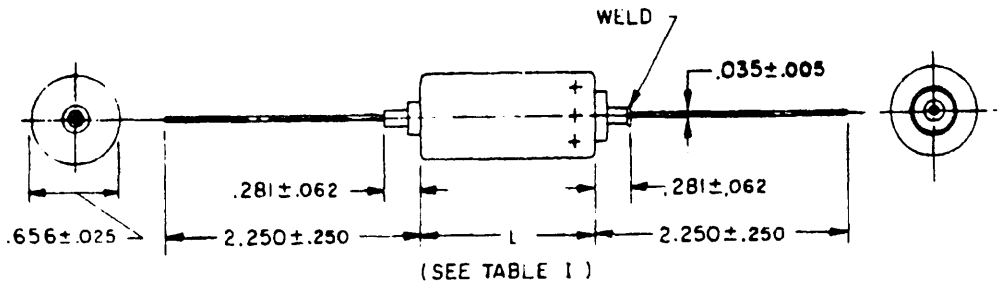
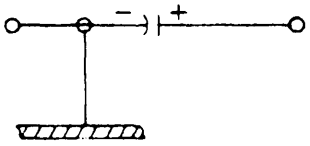
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM, (POLARIZED, SINTERED SLUG), 85°C (VOLTAGE DERATED TO 125°C), ESTABLISHED RELIABILITY, STYLE CLR10

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-39006.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.





CIRCUIT DIAGRAM

INCHES	MM
.005	.13
.025	.64
.035	.89
.062	1.57
.250	6.35
.281	7.14
.656	16.66
2.250	57.15

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Style CLR10 capacitors.

FSC 5910

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:16Z
Check the source to verify that this is the current version before use.

TABLE L Capacitor characteristics.

Rated voltage (85° C)	Cap.	DC leakage (max)			Dissipation factor (max)	Derated voltage (125° C)	Surge voltage (85° C)	Impedance (max)	Dimension L ±.062(1.57)	Part No. M39006/18- Failure rate level (%/1,000 hr)				
		25° C	85° C	125° C						L(2.0)	M(1.0)	P(0.1)	R(0.01)	S(0.001)
volts, dc	μF	μA	μA	μA	percent	volts, dc	volts, dc	ohms	inches					
8	70.0	6	30	48	47	7	9.2	60	.438 (11.13)	1000	1018	1036	1054	1072
8	140.0	10	50	80	47	7	9.2	30	.562 (14.27)	1001	1019	1037	1055	1073
10	50.0	5	25	40	35	9	11.5	75	.438 (11.13)	1002	1020	1038	1056	1074
10	100.0	9	45	72	35	9	11.5	40	.562 (14.27)	1003	1021	1039	1057	1075
20	28.0	6	30	48	21	18	23.0	85	.438 (11.13)	1004	1022	1040	1058	1076
20	56.0	10	50	80	21	18	23.0	45	.562 (14.27)	1005	1023	1041	1059	1077
30	20.0	7	35	56	15	26	34.5	125	.438 (11.13)	1006	1024	1042	1060	1078
30	40.0	12	60	96	15	26	34.5	75	.562 (14.27)	1007	1025	1043	1061	1079
60	12.0	7	35	56	9.1	52	69.0	180	.438 (11.13)	1008	1026	1044	1062	1080
60	25.0	12	60	96	9.6	52	69.0	90	.562 (14.27)	1009	1027	1045	1063	1081
90	8.0	7	35	56	6.1	78	103.0	250	.438 (11.13)	1010	1028	1046	1064	1082
90	16.0	12	60	96	6.1	78	103.0	125	.562 (14.27)	1011	1029	1047	1065	1083
180	4.0	7	35	56	6.1	155	207.0	500	.719 (18.26)	1012	1030	1048	1066	1084
180	8.0	12	60	96	6.1	155	207.0	250	.938 (23.83)	1013	1031	1049	1067	1085
270	2.5	7	35	56	5.7	235	310.0	750	1.031 (26.19)	1014	1032	1050	1068	1086
270	5.0	11	55	88	5.7	235	310.0	375	1.375 (34.93)	1015	1033	1051	1069	1087
360	2.0	7	35	56	6.1	310	414.0	1000	1.312 (33.32)	1016	1034	1052	1070	1088
360	4.0	12	60	96	6.1	310	414.0	500	1.781 (45.24)	1017	1035	1053	1071	1089

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T16:16Z
Check the source to verify that this is the current version before use.

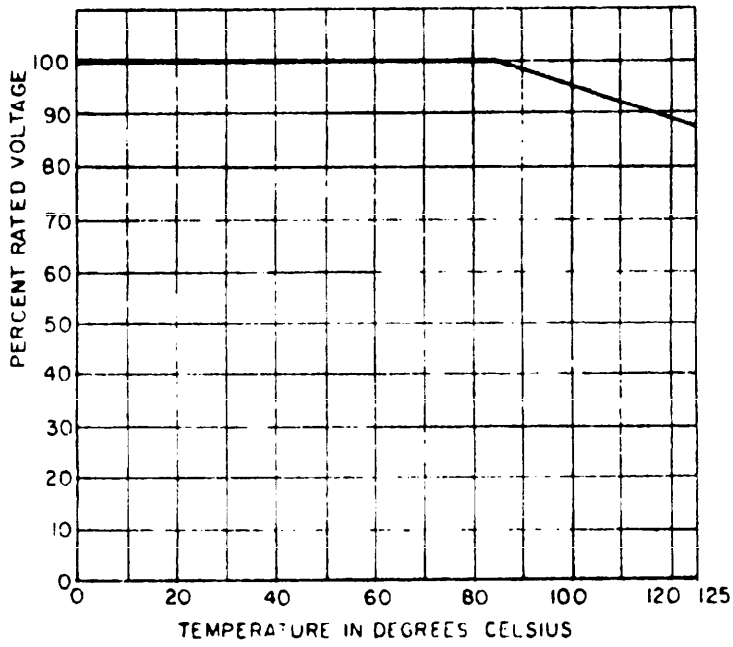


FIGURE 2. Voltage derating with temperature.

REQUIREMENTS:

Design and construction:
Dimensions and configuration - See figure 1.
Case type - Tubular, uninsulated.
Seal type - G (hermetic).
Terminals - Axial-wire lead conforming to type N3 of MIL-STD-1276, except as shown on figure 1.
Rated temperature - -55° to +85°C, voltage derated to +125°C (see figure 2).
Rated voltage - See table I.
Capacitance (Cap.) (nom) - See table I.
Cap. tolerance - +50, -15%.
FR level - See table I.

Constant voltage conditioning: As specified in MIL-C-39006.

DC leakage(DCL)(max) (at 25°C): See table I.

Cap. (nom): Method 305 of MIL-STD-202. See table I.

Dissipation factor (DF) (max): See table I.

Seal: Method 112 of MIL-STD-202, conditions A and C.

Shock (specified pulse): Method 213 of MIL-STD-202, condition I (100G).

Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20G).

Temperature cycling: Method 102 of MIL-STD-202, condition D (with step 3 at +125°C). Number of cycles - 30.
DCL - 125% (max) of 25°C value of table I.
ΔCap. - Within ±5% of initial measurement.
DF - 115% (max) of initial requirement.

Salt spray (corrosion): Method 101 of MIL-STD-202, condition B (48 hr).

Solderability: Method 208 of MIL-STD-202.

Terminal strength:
Pull - Method 211 of MIL-STD-202, condition A.
Wire-lead bend - In accordance with MIL-C-39006.

Surge voltage: See table I.

Moisture resistance: Method 106 of MIL-STD-202.
DCL - 125% (max) of 25°C value of table I.
ΔCap. - Within ±8% of initial measurement.
DF - 115% (max) of table I value.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:16Z
Check the source to verify that this is the current version before use.

Dielectric withstanding voltage (DWV):
Not applicable.

Insulation resistance (IR): Not applicable.

Low temperature (storage): Method 502 of MIL-STD-810.
DCL - See table I.
ΔCap. - Within ±5% of initial measurement.
DF - See table I.

Stability at low and high temperatures:
Step 1 (+25°C):
DCL - See table I.
Cap. - Within tolerance of table I value.
DF - See table I.
Step 2 (-55°C):
Imp. - See table I.
ΔCap. - -60% (≤20v), -40% (30v), or -30% (>30v) of step 1 value (max).
Step 3 (+25°C):
DCL - See table I.
ΔCap. - Within ±5% of step 1 value.
DF - See table I.
Step 4 (+85°C):
DCL - See table I.
ΔCap. - +30% (≤20v) or +20% (>20v) of step 1 value (max).
DF - See table I.
1/ Step 5 (+125°C):
DCL - See table I.
ΔCap. - +30% (≤18v) or +20% (>18v) of step 1 value (max).
DF - See table I.
Step 6 (+25°C):
DCL - See table I.
ΔCap. - Within ±5% of step 1 value.
DF - See table I.

Reverse voltage: Not applicable.

Life at 85°C: Method 108 of MIL-STD-202.
2,000-hr (qualification):
DCL - See table I.
ΔCap. - Within ±15% of initial measurement.
DF - See table I.
240-hr (group B):
DCL - See table I.
ΔCap. - Within ±10% of initial measurement.
DF - See table I.

10,000-hr (group C) and extended life:
DCL (85°C) - 125% (max) of table I value.
DCL (25°C) - See table I.
ΔCap. - Within ±25% of initial measurement.
DF - 200% (max) of table I value.
DWV and IR - Not applicable.

Vacuum life: Not applicable.

Life at 125°C: Method 108 of MIL-STD-202.
DCL - See table I.
ΔCap. - Within ±15% of initial measurement.
DF - 130% (max) of table I value.
DWV and IR - Not applicable.

Barometric pressure: Method 105 of MIL-STD-202, condition E (150,000 ft).
Test voltage - Rated voltage.

Fungus: Method 508 of MIL-STD-810.

Resistance to solvents: Method 215 of MIL-STD-202.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition B.
DCL - See table I.
ΔCap. - Within ±5% of initial measurement.
DF - See table I.

Marking: Capacitors shall be marked as shown in the following example:

EXAMPLE:

M39006/18-1000	- Part number
JAN (TM)	- "JAN" brand and trademark.
12345 7015A	- Source code, date code, and lot symbol.
70μF 8VDC	- Capacitance and rated voltage (at 85°C).

Polarity - Series of plus (+) symbols encircling positive end (see figure 1).

Voltage groups (85°C rated voltage):
I - - - - - - - - - - - 8 to 90v incl
II - - - - - - - - - - - 180 to 360v incl

1/ 125°C derated voltage referenced.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:16Z
Check the source to verify that this is the current version before use.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU
Navy - EC
Air Force - 11, 17, 80
DSA - ES

User activities:
Army -
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5910-0976-9)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:16Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:16Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION (Of submitter)	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

DD FORM 1426

REPLACES NAVSHIP FORM 4863, WHICH IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T16:16Z
Check the source to verify that this is the current version before use.